```java
public static int task5CP(String input1, String input2) {
    int m = input1.length();                    // Length of the first string
    int n = input2.length();                    // Length of the second string
    int[][] dp = new int[m + 1][n + 1];         // 2D array to keep track of the lengths of common subsequences
    for (int i = 1; i <= m; i++) {              // Loop through each character of the first string
        for (int j = 1; j <= n; j++) {          // Loop through each character of the second string
            if (input1.charAt(i - 1) == input2.charAt(j - 1)) {   // If the characters match
                dp[i][j] = dp[i - 1][j - 1] + 1;                  // add 1 to the length of the common subsequence
            } else {
                dp[i][j] = Math.max(dp[i - 1][j], dp[i][j - 1]);  // Otherwise, take the maximum of the previous
            }                                                     // lengths for the smaller parts of the strings
        }
    }
    return dp[m][n];                           // Return the maximum length
}

public static void main(String[] args) {
    String input1 = "abcde";
    String input2 = "ace";
    int result = task5CP(input1, input2);
    System.out.println(result);
}
```